

Khulna University of Engineering & Technology
Department of Urban & Regional Planning
BURP 1st Year 2nd Term Examination, 2019.
Math 1217
(Mathematics-II)

Full Marks: 210

Time: 3.0 hrs.

- N.B.: i) Answer any **three** questions from each section in separate script.
ii) Figures in the right margin indicate full marks.
iii) Necessary charts and graphs will be supplied.

Section – A

1. (a) Find: $\lim_{x \rightarrow 0} (1 + \sin x)^{\frac{1}{x}}$ (10)
- (b) For $f(x) = \frac{x^2 + 3x + 2}{2 - |x|}$ (15)
- I. Make a table of values of f at those values of x that approach $x_0 = -2$ from left and right. Then estimate $\lim_{x \rightarrow -2} f(x)$
- II. Also find $\lim_{x \rightarrow -2} f(x)$ algebraically.
- (c) State the Intermediate Value Theorem and using this theorem prove (10)
that the equation $\sqrt{2x + 5} = 4 - x^2$ has a solution on the interval $[0, 11]$
2. (a) A function $f(x)$ is given by $f(x) = \begin{cases} 2x - 1 & \text{when } x \geq 2 \\ 3 & \text{when } -1 \leq x < 2 \\ 1 - 2x & \text{when } x < -1 \end{cases}$ (10)
- (b) State Leibnitz's theorem. If $y = \sin(m \sin^{-1} x)$ then show that (13)
 $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} + (m^2 - n^2)y_n = 0$
- (c) Find values of the constant k and m , if possible, that will make the (12)
function f continuous everywhere, where
- $$f(x) = \begin{cases} x^2 + 5; & x > 2 \\ m(x + 1) + k; & -1 \leq x \leq 2 \\ 2x^3 + x + 7; & x \leq -1 \end{cases}$$
3. (a) Find y_n , when $y = \frac{1}{x^2 - 3x + 2}$ (10)
- (b) A garden is to be laid out in a rectangular area and protected by a (12)
chicken wire fence. What is the largest possible area of the garden if
only 100 running feet of chicken wire is available for the fence?
- (c) Find the general solution of the following differential equation: (13)
 $(D^2 - D - 2)y = e^{(e^x + 3x)}$

4. (a) Solve the initial value problem $\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 25y = 0$, $y(0) = -3$, $y'(0) = -1$ (10)
- (b) A homogeneous linear differential equation with constant coefficients has solutions $e^{-2x} \cos 3x$, $e^{2x} \sin 2x$. What are the other solution if the equation is of order four. Hence write its general solution. (10)
- (c) Find general solution of the differential equation: $(D^2 - 1)y = e^x(1 + x^2) + 5 \sin x + 10$ (15)

Section – B

5. Evaluate the following integrals:
- a) $\int \sqrt{\frac{a+x}{a-x}} dx$ (12)
- b) $\int \frac{dx}{\sqrt{(x-\alpha)(x-\beta)}}$ (12)
- c) $\int \frac{\sin x dx}{(a^2 \cos^2 x + b^2 \sin^2 x)}$ (11)
6. Evaluate the followings:
- a) $\int \sin^{-1}\left(\sqrt{\frac{x}{x+a}}\right) dx$ (11)
- b) $\int \frac{dx}{5+4 \cos x}$ (12)
- c) $\int_0^{\frac{\pi}{2}} \frac{dx}{a+b \cos x}$; $a > b$ (12)
7. (a) Evaluate $\int_0^1 \frac{x^2 \sin^{-1} x}{\sqrt{1-x^2}} dx$ (11)
- (b) Evaluate $\int_0^{\frac{\pi}{2}} \log(\sin x) dx$ (13)
- (c) Show that the area bounded by the parabolas $y^2 = ax$ and $x^2 = ay$ is $\frac{1}{3}a^2$ (11)
8. (a) Determine the general solution of the differential equation: $(2x - 3y + 4)dx + (3x - 2y + 1)dy = 0$ (12)
- (b) Write down the general form of the linear differential equation. (12)
- Solve $\frac{dy}{dx} + \frac{y}{2x} = \frac{x}{y^3}$; $y(1) = 2$
- (c) Evaluate $\int \frac{x^2+x-2}{3x^3-x^2+3x-1} dx$ (11)
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Khulna University of Engineering & Technology
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BURP 1st Year 2nd Term Regular Examination, 2019

HUM 1217
(Principles of Economics)

Full Marks: 210

Time: 3.00hrs

- N.B.** i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) How economics is like a "Science"? Explain the idea of Microeconomics and Macroeconomics. (15)
(b) What are the fundamental problems of an economy? How these problems are solved in different countries? (20)
2. (a) Define law of demand. Why the demand curve slopes downward? (05)
(b) What is market equilibrium? Find the equilibrium price and quantity of a market with demand and supply curve. (15)
(c) Solve the problem: (15)
There are 10,000 identical individuals in the market for commodity X, each with a demand equation is given by $Q_d = 100 - 6P$ and 1,000 identical producers of commodity X, each with a supply equation is given by $Q_s = 28 + 3P$
(i) Obtain equilibrium price and quantity.
Now if the government decides to collect a sales tax of \$2 per unit sold
(ii) What effect does this have on the equilibrium price and quantity?
(iii) Find the total amount of tax.
3. (a) Define price elasticity of demand. Explain the relationship between total revenue and the price elasticity of demand. (15)
(b) Solve the problems: (20)
'Bata' is a producer of leather product. The firm hires an economist to determine the demand for its product. After months of hard work the economist informed the firm's that the demand for the firms product is given by the following equation:
 $Q_B = 12,000 - 5000P_B + 5I + 500P_C$
Where,
 Q_B is the quantity demanded from Bata
 P_B , Price charged for Bata
 I , Per Capita Income
 P_C , Price of other competing producers of leather product
Assume that the initial values of P_B , I and P_C are \$5, \$10,000 and \$6, respectively, Using the above information the firm's manager wanted to determine
i) What effect a price increase would have on total revenue?
ii) How sale of the product would change during a period of rising incomes?
iii) Asses the probable impact if competing producers would raise their price?
4. (a) Define production and production function? (05)
(b) What is called factors of production? Write in short the various factors of production. (15)
(c) What is production possibilities frontiers? Explain production possibilities frontier on the basis of scarcity, choice and opportunity cost. (15)

Section – B

5. (a) What is GDP deflator? Why do economists prefer real GDP as a measure of economic well-being? Explain. (10)
(b) Discuss the methods of estimating national income. (15)
(c) What are the problems in calculating national income? (10)
6. (a) What is productivity? List and describe four determinants of productivity. (15)
(b) What is inflation? Explain the idea of demand pull inflation and cost push inflation (20)
7. (a) What is the role of the financial system? Name and describe two markets that are financial system of Bangladesh. (15)
(b) What is private saving and public saving? Suppose GDP is \$ 8 trillion, taxes are \$ 1.5 trillion, Private savings is \$ 0.5 trillion and public saving is \$ 0.2 trillion. Assume this economy is closed; calculate consumption, government purchase, national savings and investment. (20)
8. (a) Describe the main function of a central bank. How far does the Bangladesh Bank perform those functions? (15)
(b) What are the objectives of Linear Programming? Suppose, A specialist steel manufacturer produces two types of steel g_1 and g_2 . Type 1 requires 2 hours of melting, 4 hours of rolling, and 10 hours of cutting. Type 2 requires 5 hours of melting, 1 hour of rolling, and 5 hours of cutting. Forty hours are available for melting, 20 for rolling, and 60 for cutting. The profit margin for type 1 is 24; For type 2 it is 8. Reduce the data to the equations and inequalities necessary to determine the output mix that will maximize profit. (20)

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BURP 1st Year 2nd Term Regular Examination, 2019
URP 1251
(Introduction to Architectural History)

Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) What is architectural heritage? (05)
(b) What are the social, technological and cultural influences on architecture in different eras? (10)
(c) What are the architectural characteristics of Egyptian Architecture? (10)
(d) Draw the Cross-section of "The Great Pyramid of Cheops (Khufu)". (10)
2. (a) Draw the master plan of "The Greek Agora" or "The Forum Romanum". (15)
(b) Explain Greek Order and Optical Corrections. (20)
3. (a) Draw the graphical details of "The Pantheon". (15)
(b) Explain with necessary sketches: (20)
 - (i) Ribbed Vault
 - (ii) Nave
 - (iii) Cheveltes
 - (iv) Clerestory
 - (v) Buttress
4. (a) What are the main architectural characteristics of Renaissance Architecture? (10)
(b) What were the causes of failure of Modern Movement? (10)
(c) Write about the following topics: (15)
 - (i) New York Five
 - (ii) Cubism
 - (iii) Deconstructive Architecture

Section – B

5. (a) Briefly explain the development of Buddhist Architecture in Bengal. (20)
(b) State the architectural characteristics of traditional style of temple architecture with necessary sketches. (10)
(c) Explain the concept of "Shrine" of Buddhist Architecture. (05)
6. (a) • What do you mean by "Ethical Brutal Approach"? (10)
(b) What is the nexus between Modern Architecture and International Style? (10)
(c) Explain the concept of "Art Noveau". Describe the link between Cubism and Expressionism. (15)
7. (a) How did Khan-e-Jahan Style influence Bengal Architecture? (10)
(b) Briefly explain the architectural characteristics of Khan-E-Jahan Style. (10)
(c) "Curzon hall is the blend of European and Mughal Style"- Explain the statement with your logical arguments. (15)
8. (a) Write short notes (Give example and sketch where applicable): (35)
 - (i) Development of Bungalow
 - (ii) Classification of houses
 - (iii) Symmetrical composition of Colonial Public Approach